INCH-POUND

MIL-M-63320A(AR)
AMENDMENT 3
27 MARCH 1989
SUPERSEDING
AMENDMENT 2
19 November 1982

MILITARY SPECIFICATION

MICROCIRCUIT, DIGITAL, CMOS (INITIAL LOGIC)

This amendment forms a part of Military Specification MIL-M-63320A(AR) dated 8 April 1981, and is approved for use within the US Army Armament Munitions and Chemical Command and is available for use by all the Departments and Agencies of the Department of Defense.

PAGE 3

3.4, line 1: Delete "Environmental tests" and substitute "Mechanical tests".

Mechanical shock, Conditions: Delete "D" and substitute "C".

* Delete "Method 2015.1" and substitute "Method 2015".

* (This change reinstates original provision of specification.)

3.5.4., (a) and (b): Delete (-45°C to +85°C) and substitute (-40°C to +70°C).

PAGE 7

4.3.1 Submission: Delete last sentence and substitute "The first article sample, after screening, shall consist of the quantities needed for double samplings stipulated in Table III."

PAGE 8

4.3.4: Add "e. A lapse in production for a period exceeding 90 days or as stipulated in the contract".

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TABLE III, Subgroups a, b, c, d: Delete "25" and substitute "8/".

Subgroup b, Mechanical shock: Delete condition "D" and substitute condition "C".

* Subgroup c, Delete "Method 2015.1" and substitute "Method 2015".

* (This change reinstates original provision of specification.)

TABLE III, Subgroup g: Delete "25" and substitute "9/".

NOTES: Delete present content of note 1 and substitute "Devices which fail to any test criteria in the screening sequence shall be removed from the first article sample. The contractor should choose a first article sample that will provide, after screening, the quantities stipulated in notes 8 and 9".

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TABLE III: Add the following notes:

"8/ A double sampling plan shall be used as follows: 25-0-2, 38-1-2. For group b replace fallouts from subgroup a.

9/ A double sampling plan shall be used as follows: 45-0-2, 77-1-2."

PAGE 14

4.4.2.2, AQL or 100% Column: Delete "All LTPD's" and substitute "LTPD's of 10%, 10%, 10% and 10%".

PAGE 15

* 4.4.2.3, Subgroup 2: Delete "Method 2015.1" and substitute "Method 2015".

(This change reinstates original provision of specification.)

Delete "Bond strength (bond pulls 1/), ---, LTPD 15%". Substitute "Bond strength (bond pulls 1/), 5, 1-2".

Delete Notes "1/" in its entirety and substitute: 1/ Pull all bond wires on each device - reject for two (2) or more bond failures."

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4.4.2.4, Subgroup 2, Mechanical shock: Change Condition "D" to Condition "C".

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* 4.5.17, Burn-in screen, line 6: Delete "7.5 volts dc" and substitute "8.0 volts dc".

* 4.5.18, Steady state life, line 3: Delete:

"Test duration shall be 100 hours minimum at 125°C ± 5°C"

And substitute:

"Test duration shall be 1000 hours minimum at 125°C ± 5°C or equivalent per Method 1005 of MIL-STD-883B. Temperature in excess of 135°C + 8°C and Test duration less than 496 hours shall not be allowed.".

PAGE 21

6.3 and 6.5: Delete "QAA" and substitute "QAT".

PAGE 23

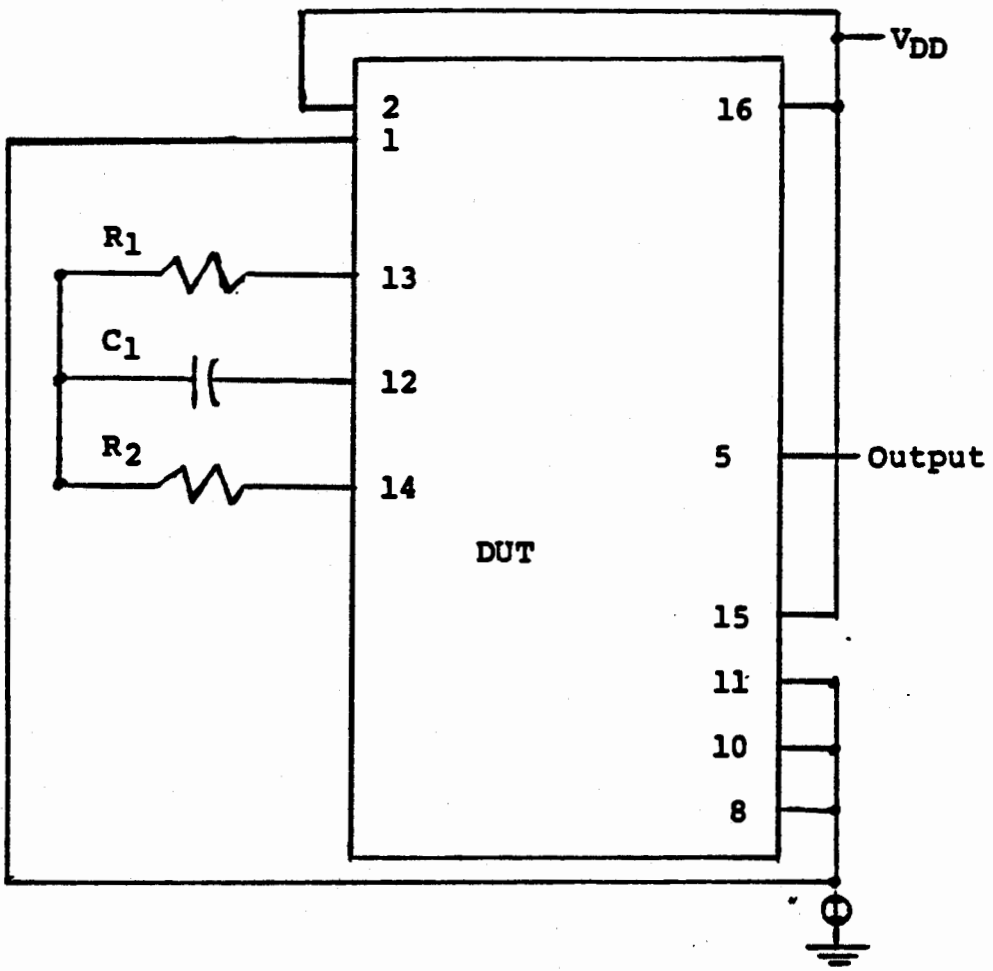
FIGURE 1: Delete present and substitute revised FIGURE 1 included to this amendment.

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FIGURE 15: Delete present and substitute revised FIGURE 15 included to this amendment.

* FIGURE 15: Delete "7.5 ± 0.1 V dc, + 7.5 ± 0.1 V (Square Wave) and substitute "+8.0 ± 0.5 V dc, + 8.0 ± 0.5 V (Square Wave)".

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	NOMINAL	TOLERANCE	STYLE	TYPE	TCC
C1	0.01 uf	± 1.0%	Ceramic	NPO	± 25 ppm
R1	2.0 M ohms	± 5.0%	Composition		
R2	470 K ohms	± 1.0%	Metal Film		± 50 ppm

FIGURE 1. Test Circuit for Leakage Current (Oscillator) and Oscillator Function tests.

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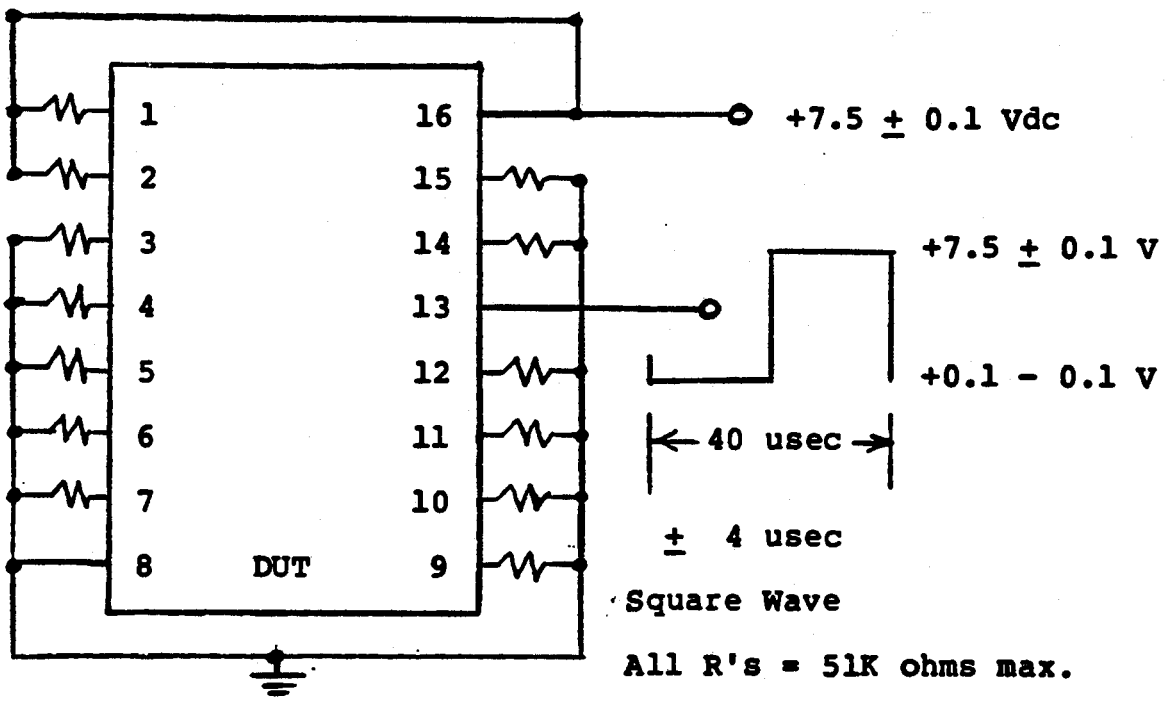


FIGURE 15. Burn-in and Steady State Life-Test Circuit.

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